

14_Indefinite Matrix | Convex & Non-Convex Function

1. Indefinite Matrix

A **real symmetric matrix** A is called **indefinite** if its associated quadratic form

$$x^T A x$$

takes **both positive and negative values** for different non-zero vectors x .

Classification (for symmetric matrices)

- **Positive definite:** $x^T A x > 0$ for all $x \neq 0$
- **Negative definite:** $x^T A x < 0$ for all $x \neq 0$
- **Positive semidefinite:** $x^T A x \geq 0$
- **Negative semidefinite:** $x^T A x \leq 0$
- **Indefinite:** $x^T A x$ is sometimes positive, sometimes negative

Eigenvalue test

For a **symmetric matrix**:

- All eigenvalues $> 0 \rightarrow$ positive definite
- All eigenvalues $< 0 \rightarrow$ negative definite
- **Some eigenvalues > 0 and some $< 0 \rightarrow$ indefinite**

Example

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Eigenvalues: $1, -1$
 So A is **indefinite**.

$$A = \begin{bmatrix} 2 & 4 \\ 4 & -3 \end{bmatrix}$$

$$\begin{vmatrix} 2 - \lambda & 4 \\ 4 & -3 - \lambda \end{vmatrix} = 0$$

$$(2 - \lambda)(-3 - \lambda) - (4)(4) = 0$$

$$-6 - 2\lambda + 3\lambda + \lambda^2 - 16 = 0$$

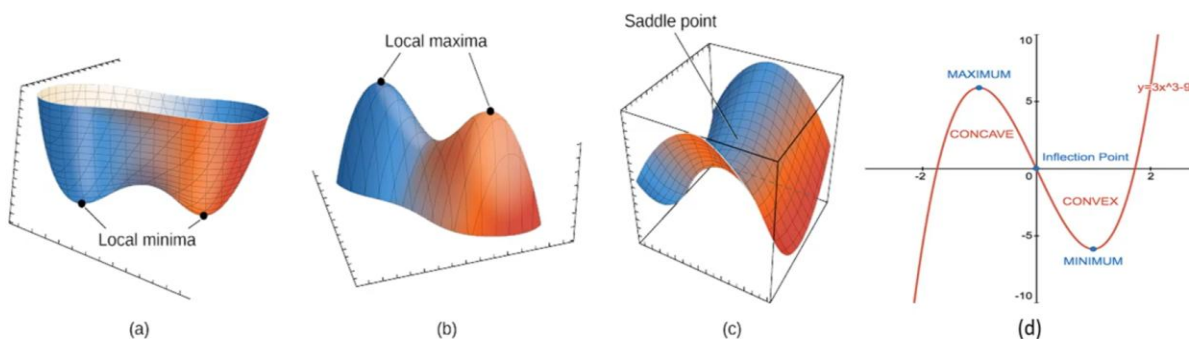
$$\lambda^2 + \lambda - 22 = 0$$

$$\lambda_1 \approx \frac{-1 + 9.43}{2} = \frac{8.43}{2} = 4.215$$

$$\lambda_2 \approx \frac{-1 - 9.43}{2} = \frac{-10.43}{2} = -5.215$$

Since one eigenvalue is **positive** and the other is **negative**, the matrix is **indefinite**.

2. Convex and Non-Convex Functions



- Convex (Minimum) → Hessian is Positive Definite ($\lambda > 0$)
- Concave (Maximum) → Hessian is Negative Definite ($\lambda < 0$)
- Saddle Point → Hessian is Indefinite (some $\lambda > 0$, some $\lambda < 0$)

Convex Function

A function $f(x)$ is **convex** if for all x, y and $\lambda \in [0, 1]$:

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

Geometric meaning:

The line segment between any two points on the graph lies **above** the graph.

Non-Convex Function

If the above inequality does **not** hold, the function is **non-convex**.

3. Connection Between Matrices and Convexity

Consider a **quadratic function**:

$$f(x) = x^T A x$$

where A is a symmetric matrix.

Convexity depends on A :

- **A positive semidefinite** $\rightarrow f(x)$ is **convex**
- **A positive definite** $\rightarrow f(x)$ is **strictly convex**
- **A indefinite** $\rightarrow f(x)$ is **non-convex**

Why?

If A is indefinite, the function curves **upward in some directions** and **downward in others**, creating saddle points.